[ENTER COMPANY NAME]

QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Process and Equipment Validation Framework

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Current Version / Revision	00
Effective Date	[Date of Final Approval]
Review Cycle	Annual (Every 12 Months)
Applicable Regulatory Standards	ISO 13485:2016 Clauses 7.5.6 & 7.5.7 FDA 21 CFR Part 820.75 GHTF/SG3/N99-10 (Process Validation Guidance)

Prepared By	Reviewed By	Approved By
Name: [Enter Name] **Title:** Validation Engineer	**Name:** [Enter Name] **Title:** Manufacturing Manager	**Name:** [Enter Name] **Title:** Head of Quality Operations
Date: ______________	**Date:** ______________	**Date:** ______________

1.0 Purpose

This Standard Operating Procedure (SOP) defines the mandatory requirements, structural phases, and objective success metrics for executing process, utility, and equipment software validations across all manufacturing configurations. The principal objective is to provide high-degree statistical assurance that a specialized production process will consistently yield devices meeting pre-established material specifications, operational capabilities, and safety properties.

2.0 Scope

This procedure applies to all automated manufacturing equipment, software tracking layers, critical cleanroom infrastructure utilities, inspection scripts, and specialized processes where the physical output cannot be fully verified by subsequent non-destructive terminal testing (e.g., ultrasonic weld joins, heat sealing, chemical washing, and molding cycles).

3.0 Responsibility and Authority

3.1 Quality Assurance / Validation Engineer

Responsible for drafting formal Validation Master Plans (VMP), populating prospective execution protocols (IQ/OQ/PQ), analyzing technical multi-variable datasets, and compiling final validation reports.

3.2 Manufacturing Operations / Equipment Owners

Responsible for establishing hardware parameter boundaries, maintaining cleanroom tooling interfaces, ensuring preventive maintenance calibrations are current, and running test batches during validation windows.

3.3 Validation Review Board (VRB)

A multi-functional compliance panel responsible for verifying protocol challenge setups, confirming statistical sample sizes, enforcing technical deadlines, and signing off on the commercial activation log.

4.0 Procedure

4.1 Process Validation Mandate Assessment Matrix

Engineering teams must evaluate every production step to determine if process validation is mandatory or if standard inspection suffices. The decision pathway follows GHTF guidance:

Process Type Profile	Technical Characterization	Mandated Quality Control Path
Fully Verifiable Output	The resulting parameter can be 100% non-destructively checked (e.g., automated laser dimensional logging).	Standard inspection controls + equipment calibration checks.
Destructive / Non-Verifiable	The parameter cannot be verified without destroying the device, or testing is economically unfeasible (e.g., sterile sealing pouch welds).	**Full Validation Mandate** (Complete IQ/OQ/PQ lifecycle execution).
Software/Automated Layer	Production tracking software, logic controllers, or automated vision inspection loops.	**Software Validation** (Aligned with FDA Part 11 and GAMP 5 models).

4.2 Aligned Validation Phases Lifecycle

When formal process validation is triggered, the execution must progress through distinct, sequential phases managed under strict change controls:

Phase 1: Installation Qualification (IQ)
Confirm that the equipment, utility plumbing, and data connections are installed and hooked up according to design prints. Collect physical records of calibration settings, safety interlock checks, software version levels, and preventative maintenance logs.

Phase 2: Operational Qualification (OQ)
Evaluate the process capabilities across its full operating boundaries. Challenge the system at its defined upper and lower limits (worst-case settings, such as low temperature combined with high feed rate). Execute fractional factorial Design of Experiments (DOE) to isolate critical process parameters (CPPs) and define operational windows.

Phase 3: Performance Qualification (PQ)
Demonstrate that the process operates consistently under normal manufacturing conditions over an extended period. PQ runs require processing multiple consecutive production lots (a minimum of three successful, consecutive batches is standard) using real raw material lots and normal operator shifts.

> **TECHNICAL COMPLIANCE NOTE:** Any deviation, mechanical failure, or software error occurring during an active OQ or PQ run triggers a mandatory investigation. The validation run must be halted, a deviation report logged, and the specific failure resolved before the consecutive batch count can restart from zero.

4.3 Statistical Sample Size Selection and Acceptance Targets

Validation protocols must justify their selected sample sizes using established statistical risk criteria. For variable engineering data (e.g., tensile strength, weld energy), sample sizes must demonstrate a minimum Process Capability Index (C_{pk}) target of ≥ 1.33 for standard operations, and ≥ 1.66 for critical safety features. Attribute data validation (pass/fail checkmarks) requires specific confidence and reliability thresholds (e.g., 95% Confidence / 99% Reliability boundaries) calculated using the binomial distribution matrix.

4.4 Validation Re-Trigger Parameters

Once a process state is approved and signed off by the VRB, it enters routine monitoring status. A full or partial re-validation protocol must be evaluated if any of the following modification limits are crossed:

- Shifting hardware fixtures or moving heavy automated lines to new cleanroom facility pads.
- Substantial raw material changes or swapping out primary baseline component suppliers.
- Recurring nonconformity profiles or negative post-market complaint trends targeting specific process steps.
- Completing major restorative maintenance or upgrading core PLC operating software.

5.0 Documentation Retention

All approved Validation Master Plans, individual protocols (IQ/OQ/PQ), raw measurement sheets, statistical charts, deviations, and finalized validation closure reports are critical quality assets. They must be maintained securely for the active lifecycle of the device family plus a minimum of ten (10) years, ensuring they are readily available for regulatory audit teams.

6.0 Associated Reference Materials

- **SOP-QA-001:** Document and Data Control System
- **SOP-QA-002:** Change Control System
- **SOP-MF-010:** Sterilization Process Management and Validation
- **SOP-MF-011:** Contamination Control and Product Cleaning [cite: 19]

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], Quality Engineering	Initial baseline release of formalized Process and Equipment Validation Framework Standard Operating Procedure.